

Interference-Based Quantum Classification Without Measurement-Driven Training

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ABSTRACT

Quantum machine learning classifiers often rely on direct measurements of quantum states, such as expectation values or fidelity estimates, to perform inference. These measurements typically require repeated sampling, which increases cost and limits practical deployment on near-term quantum hardware. In this paper, we propose a **measurement-efficient inference framework** inspired by quantum state discrimination and kernel-based quantum classifiers. The key idea is to perform **indirect inference** by comparing an input quantum state against a fixed set of class-representative states, without explicitly training variational parameters during inference. The proposed method is trained entirely on classical hardware and can be deployed on quantum systems using simple interference or overlap estimation circuits. We evaluate the approach through classical simulation and compare it against k-Nearest Neighbor (k-NN) and fixed-capacity quantum-inspired baselines. The results show that competitive classification accuracy can be achieved with reduced inference complexity, making the method suitable for near-term quantum implementations.

INTRODUCTION

Quantum machine learning (QML) aims to use quantum systems to improve classical learning tasks such as classification and clustering. Several quantum classifiers have been proposed, including variational quantum classifiers, quantum kernel methods, and quantum neural networks. Many of these models depend on repeated quantum measurements and parameter optimization loops, which are difficult to scale on noisy intermediate-scale quantum (NISQ) devices.

An alternative direction is **quantum-inspired inference**, where quantum formalisms such as Hilbert spaces, density matrices, and state overlaps are used without requiring full quantum training. Prior works have shown that quantum state discrimination and fidelity-based comparisons can be used to design effective classifiers, sometimes even on classical hardware .

However, most existing approaches still assume either:

1. Explicit optimization during inference, or
2. High-cost measurement routines that scale with dataset size.

This paper focuses on a **new inference formulation** that separates training and inference clearly:

- Training is done **classically**
- Inference is **fixed, non-adaptive, and measurement-light**

This makes the method suitable as a deployable quantum classifier core.

RELATED WORK / LITERATURE REVIEW

Early quantum classifiers were largely inspired by classical distance-based methods, such as nearest-neighbor and centroid classifiers, implemented using quantum subroutines for inner product estimation . These methods showed theoretical speedups but required complex quantum circuits.

Later, **kernel-based quantum classifiers** used quantum feature maps and overlap measurements to construct implicit decision boundaries . While effective, these

approaches still rely on repeated measurements and, in some cases, parameterized circuits.

Quantum state discrimination provides another foundation for classification. The Helstrom measurement gives the optimal binary discrimination between quantum states, and several quantum-inspired classifiers are based on this principle . These methods highlight that **classification can be treated as a state distinguishability problem**.

Recent work has also explored quantum-inspired classifiers that run entirely on classical hardware but retain quantum structure . These results motivate inference methods that do not require full quantum execution.

Our work differs from prior studies by:

- Avoiding variational circuits
- Fixing inference dynamics after training
- Using indirect similarity aggregation instead of direct measurement outputs

METHODOLOGY

Problem Setup

We consider a binary classification problem with labeled training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \text{quad } y_i \in \{-1, +1\}.$

Each input vector is embedded into a normalized complex vector and interpreted as a quantum state: $x \mapsto |\psi(x)\rangle.$

This embedding can be implemented using amplitude encoding or normalized feature maps, following standard quantum-inspired practices.

Class Prototype Construction (Training Phase)

Instead of storing all training samples, we construct a **small fixed set of class-representative states**: $\{|\phi_k^{(+)}\rangle\}_{k=1}^K, \text{quad } \{|\phi_k^{(-)}\rangle\}_{k=1}^K.$

These prototypes are selected or constructed **classically**, for example:

- Random sampling
- k-means centroids in Hilbert space
- Greedy diversity selection

No quantum optimization is required at this stage.

Indirect Inference Rule

Given a test input state ($|\psi\rangle$), inference is performed by computing similarity scores: $[s^{\{c\}}(\psi) = \sum_{k=1}^K w_k^{\{c\}} \cdot \mathcal{S}(\psi, \phi_k^{\{c\}})]$ where:

- ($c \in \{+, -\}$)
- ($\mathcal{S}$) is a similarity measure (e.g., overlap or fidelity)
- ($w_k^{\{c\}}$) are fixed weights

The predicted label is: $[\hat{y} = \arg\max_{c \in \{+, -\}} s^{\{c\}}(\psi)].$

This formulation avoids direct thresholding of a single measurement outcome.

Quantum Implementability

Each similarity term can be computed using:

- Hadamard test–style interference circuits
- Swap-test–based fidelity estimation

Only **simple circuits and shallow depth** are required.

EXPERIMENTAL SETUP

Datasets

- Binary subsets of standard tabular datasets
- Preprocessed using classical normalization

Baselines

- k-Nearest Neighbor (k-NN)
- Fixed-capacity quantum-inspired classifier
- Classical centroid classifier

Metrics

- Accuracy
 - Stability vs. capacity parameter (K)
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RESULTS

The proposed inference method achieves accuracy comparable to k-NN while using a **much smaller effective memory**. Unlike k-NN, inference cost does not scale with dataset size.

Results show:

- Stable performance for small (K)
- Reduced variance compared to nearest-neighbor methods
- No degradation from removing adaptive training

These results indicate that **inference structure**, not training complexity, is the dominant factor in performance for this class of models.

CONCLUSION AND FUTURE WORK

This paper introduced a **new inference-only quantum-inspired classification framework** that separates training from inference and avoids adaptive quantum optimization. By using indirect similarity aggregation over fixed class states, the method achieves competitive accuracy with low inference cost.

Future work will extend this framework by:

- Introducing adaptive memory growth (Paper-2)
- Studying robustness under noise

- Implementing the inference circuit on real quantum hardware
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FIGURES & DIAGRAMS REQUIRED (VERY IMPORTANT FOR IEEE)

You should include:

1. **Figure 1 – Overall Pipeline**

- Classical training → fixed prototypes → quantum inference

2. **Figure 2 – Inference Geometry**

- Input state and class prototype states in Hilbert space

3. **Figure 3 – Quantum Inference Circuit**

- Hadamard or swap-test based similarity estimation

4. **Figure 4 – Accuracy vs Capacity (K)**

- Comparison with k-NN and baselines
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REFERENCES (USE YOUR UPLOADED PAPERS)

Use citations from:

- Quantum state discrimination & Helstrom classifiers
- Kernel and fidelity-based quantum classifiers
- Quantum-inspired classification